IE2024 - Probability & Statistics

Probability Theory and Distributions

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MSc, MBA, PGDip, BSc (1st Class)



Experiment

- In probability theory, an experiment is a process or action that generates a set of possible outcomes.
- Each time the experiment is performed, it can result in any one of these outcomes.
  - Flipping a coin.
  - Rolling a Die
  - Drawing a Card from a Deck

Trial

- A trial is a single occurrence or realization of a probability experiment.
- It is an individual instance in which the experiment is conducted, and an outcome is observed. Examples are,
  - Flipping the coin once
  - Rolling the die once

Sample Space

- The sample space (denoted as S) is the set of all possible outcomes of a probability experiment.
- Each individual outcome in the sample space is called a sample point or elementary event.
- The sample space provides a complete description of all the potential results that can occur in an experiment.

Events

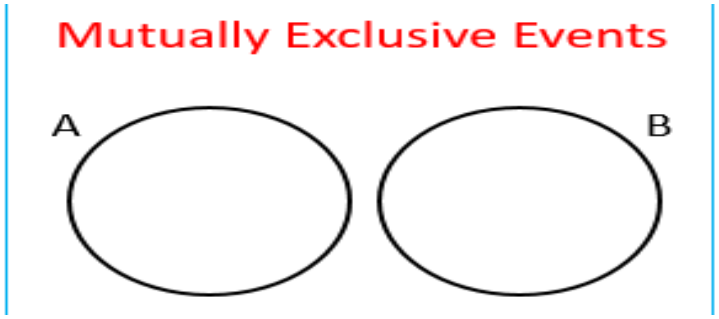
- An event is a subset of the sample space
  - Event of getting at least one head when two coins are tossed.
  - {(HH), (HT), (TH)}
- Simple event : An event which has a single possible outcome
- Compound event : An event with multiple outcomes

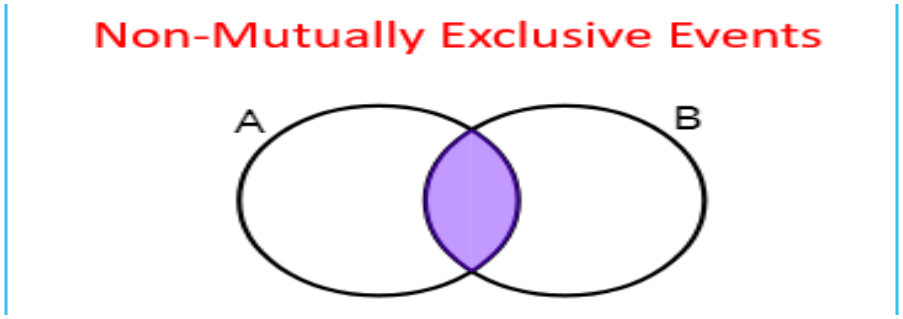
Events

- The null subset ($\emptyset$) of S is called an impossible event.
  - The event $A \cup B$ consists of all outcomes that are in A or in B or in both.
  - The event $A \cap B$ consists of outcomes that are both in A and B.
  - The event A^c(the complement of A in S) consists of all outcomes not in A, but in S.

Mutually Exclusive Events

- Mutually exclusive events are events that cannot occur simultaneously.
- In other words, if one event occurs, the other event cannot occur at the same time.
- These events have no outcomes in common.
- Two events A and B are mutually exclusive if $A \cap B = \emptyset$, meaning their intersection is the empty set.





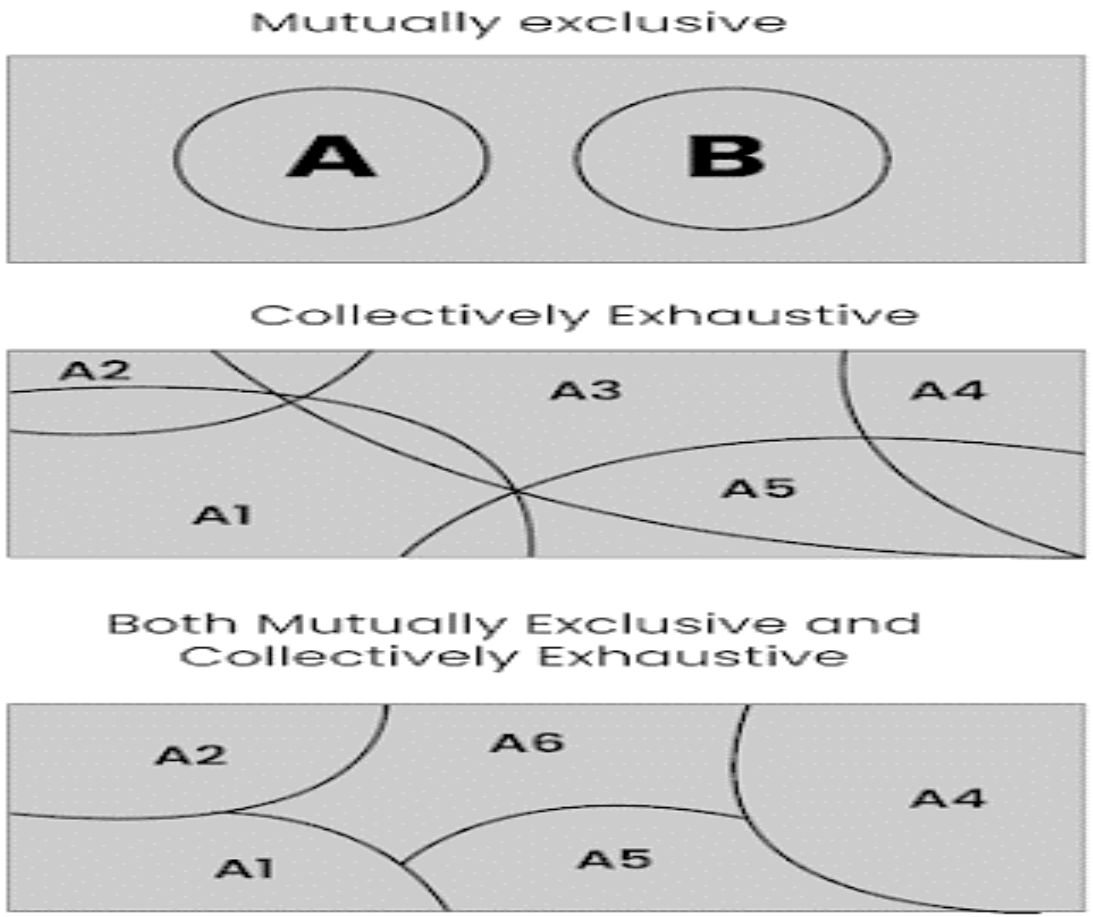
Mutually Exclusive Events

- Examples:
  - Flipping a coin
    - A: Getting heads.
    - B: Getting tails.
  - Rolling a six-sided die
    - A: Rolling an even number (2, 4, 6).
    - B: Rolling an odd number (1, 3, 5).
  - Drawing a single card from a standard deck
    - A: Drawing a red card (hearts or diamonds).
    - B: Drawing a black card (spades or clubs).

Collectively Exhaustive Events

- Collectively exhaustive events are a set of events that cover all possible outcomes of an experiment.
- In other words, at least one of the events must occur when the experiment is performed.
- For a set of events $A1, A2, \ldots, An$ to be collectively exhaustive in a sample space S : $A1 \cup A2 \cup \cdots \cup An = S$

Collectively Exhaustive Events



Collectively Exhaustive Events

- Examples:
  - Rolling a six-sided die
    - A 1 : Rolling a 1.
    - A 2 : Rolling a 2.
    - A 3 : Rolling a 3.
    - A 4 : Rolling a 4.
    - A 5 : Rolling a 5.
    - A 6 : Rolling a 6.
  - Flipping a coin
    - A: Getting heads.
    - B: Getting tails.

Independent Events

- In probability theory, independent events are events whose occurrence or non-occurrence does not affect the occurrence of the other event.
- In other words, two events A and B are independent if the occurrence of A does not change the probability of B occurring, and vice versa.
- It's common to find the concepts of independent events and mutually exclusive events confusing
  - Independence - The occurrence of one event does not affect the probability of the other event.
  - Mutual Exclusivity - The occurrence of one event means the other event cannot occur.

Independent Events

- Examples:
  - Flipping a coin twice.
    - A: Getting heads on the first flip.
    - B: Getting heads on the second flip.
  - Rolling a die and flipping a coin.
    - A: Rolling a 4.
    - B: Getting tails on the coin flip.
  - Drawing a card from a deck, replacing it, and drawing again.
    - A: Drawing an Ace on the first draw.
    - B: Drawing a King on the second draw.

Examples

- Consider a deck of cards. Let A – Aces, B – Black cards, C – Diamonds, and D – Hearts. Find collectively exhaustive events and mutually exclusive events.
- Let E – Even-numbered cards, F – Face cards, G – Spades, and H – Clubs. Find collectively exhaustive events and mutually exclusive events.

Probability

- Probability is a branch of mathematics that deals with the likelihood of different outcomes occurring in uncertain situations.
- It quantifies how likely an event is to happen, with values ranging from 0 to 1, where:
  - 0 represents an event that is impossible. (100% sure of non-occurrence)
  - 1 represents an event that is certain. (100% sure of occurrence)
  - Values between 0 and 1 represent the likelihood of the event occurring.
  - The probability of the sample space S is always 1 ($P(S)=1$).
  - The probability of an impossible event is always 0 ($P(\emptyset)=0$).

Classical Definition of Probability

- The probability of an event E occurring is defined as the ratio of the number of favorable outcomes to the total number of possible outcomes in the sample space, assuming each outcome is equally likely.

$$P(E) = \frac{n(E)}{n(S)}$$

- The classical definition of probability is based on the idea of equally likely outcomes.
- It is used when all outcomes of an experiment are equally probable.

Examples

- A balanced/fair die (with all outcomes equally likely) is rolled. Let A be the event that an even number occurs. What is the probability of A?

$$P(A) = \frac{3}{6} = 0.5$$

- Suppose we toss two coins. Assume that all the outcomes are equally likely (fair coins). Let A be the event that at least one of the coins shows up heads. Find $P(A)$.

$$P(A) = \frac{3}{4} = 0.75$$

Basic Properties of Probability

- $P(A^c)=1-P(A)$
- $P(A \cup B)=P(A)+P(B)-P(A \cap B)$
- If $A \cap B=\emptyset$ (i.e., A and B are mutually exclusive),
  - $P(A \cup B)=P(A)+P(B)$
- If $A1, A2, \ldots, Ak$ are mutually exclusive events,
  - $P(A1 \cup A2 \cup \cdots \cup Ak)=P(A1)+P(A2)+\cdots+P(Ak)$

Examples

- In a large university, the freshman profile for one year's fall admissions indicates that 40% of the students were in the top 10% of their high school class, and 65% are white. Additionally, 25% of the students were both white and in the top 10% of their high school class. What is the probability that a freshman student selected randomly from this class was either in the top 10% of their high school class or is white?

Examples

- Let:
  - A = Student is in the top 10% of their high school class
  - B = Student is white
- Given:
  - P(A) = 0.40
  - P(B) = 0.65
  - P(A ∩ B) = 0.25

Examples

- Using the **Addition Rule of Probability**:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- Substitute the given values:

$$P(A \cup B) = 0.40 + 0.65 - 0.25$$
$$P(A \cup B) = 0.80$$

Joint Probability

- The probability of events A and B occurring together is defined as the joint probability of A and B.

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}$$

Examples

- Find the probability of drawing a Black Ace from a standard deck of playing cards if a card is drawn at random.
- Find the probability of drawing a Red Jack from a standard deck of playing cards if a card is drawn at random.

Question 1: Probability of Drawing a Black Ace

- A standard deck of cards contains:
  - 52 cards
  - 2 Black Aces (Ace of Spades ♠ and Ace of Clubs ♣)
- Using the probability formula:

$$P(\text{Black and Ace}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

$$P(\text{Black Ace}) = \frac{2}{52}$$

$$P(\text{Black Ace}) = \frac{1}{26}$$

Question 2: Probability of Drawing a Red Jack

- A standard deck of cards contains:
  - 52 cards
  - 2 Red Jacks (Jack of Hearts ♥ and Jack of Diamonds ♦)
- Using the probability formula:

$$P(\text{Red and Jack}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

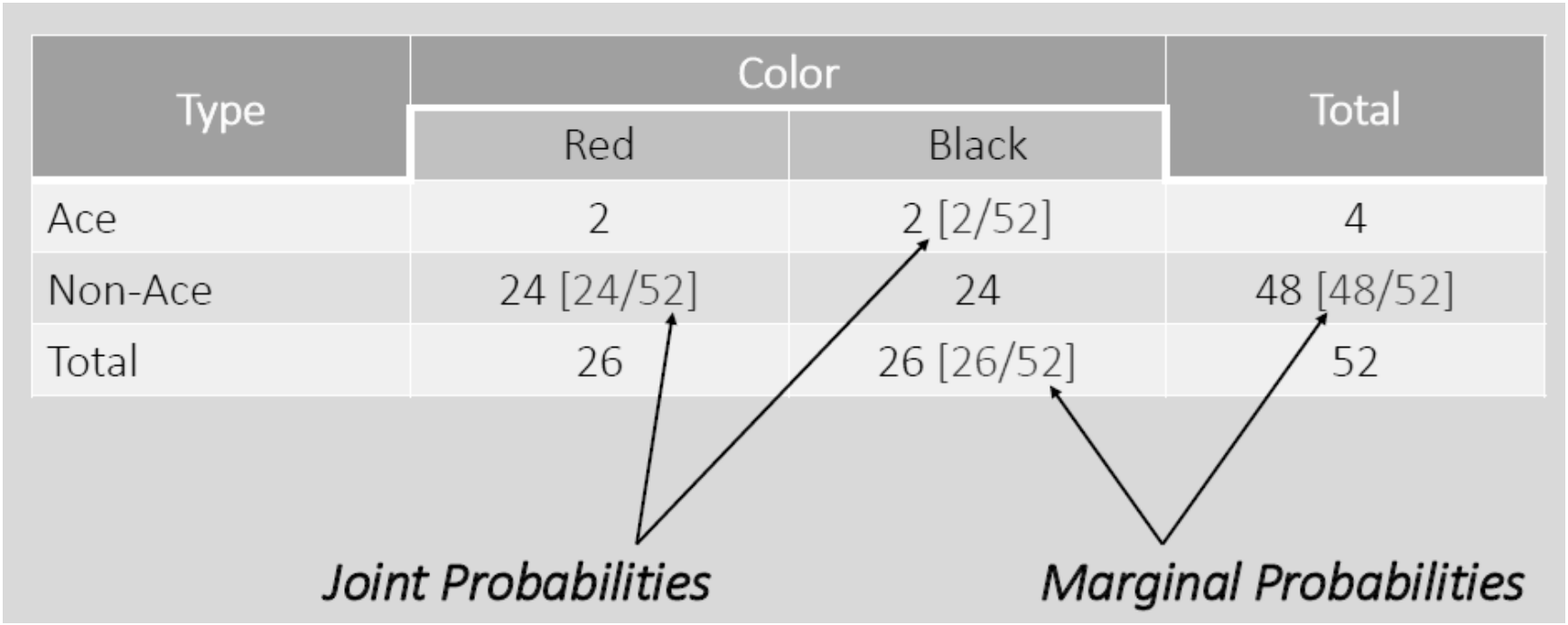
$$P(\text{Red Jack}) = \frac{2}{52}$$

$$P(\text{Red Jack}) = \frac{1}{26}$$

Examples

- Find the probability of drawing a Black Ace from a standard deck of playing cards if a card is drawn at random.

Type	Color		Total
	Red	Black	
Ace	2	2 [2/52]	4
Non-Ace	24 [24/52]	24	48 [48/52]
Total	26	26 [26/52]	52



Conditional Probability

- This is the probability of one event, given that another event has already occurred.
- The conditional probability of an event A, given that an event B has already occurred, is denoted by $P(A|B)$.

$$\boldsymbol{Pr(A \mid B) = \frac{Pr(A \cap B)}{Pr(B)}} \quad ; Pr(B)>0$$

$$\boldsymbol{Pr(B \mid A) = \frac{Pr(A \cap B)}{Pr(A)}} \quad ; Pr(A)>0$$

;Where $P(A \cap B)$ = Joint probability of A and B

$P(A)$ = Marginal probability of A

$P(B)$ = Marginal probability of B

Examples

- Of the cars on a used car lot, 70% have air conditioning and 40% have a CD player. Additionally, 20% of the cars have both features. What is the probability that a car has a CD player, given that it has air conditioning?
- If two balanced dice are tossed, find the probability that the sum of the face values is 8, given that the face value of the first die is 3.

Question 01

- Let:
  - A = Car has air conditioning
  - B = Car has a CD player
- Given:
  - P(A) = 0.70
  - P(B) = 0.40
  - P(A ∩ B) = 0.20

Question 01

- Using the Conditional Probability Formula:
  - P(B|A)=P(A∩B) / P(A)
- Substitute the given values:
  - $P(B \mid A) = \frac{0.20}{0.70}$
  - $P(B \mid A) = 0.2857$

Question 02

- Let:
  - A = The sum of the two dice is 8
  - B = The first die shows 3
- We are asked to find:
  - $P(A \mid B)$

Question 02

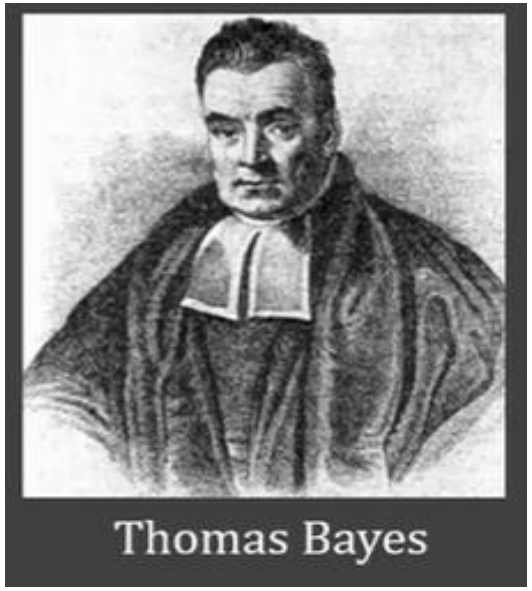
- Using the Conditional Probability Formula:
  - $P(A|B)=P(A \cap B) / P(B)$
- Since the first die is already known to be 3, the possible outcomes are:
  - (3,1)
  - (3,2)
  - (3,3)
  - (3,4)
  - (3,5)
  - (3,6)
- There are 6 possible outcomes.

Question 02

- To obtain a sum of 8, the second die must show 5.
- Favorable outcome:
  - (3,5)
- Number of favorable outcomes = 1
- Total possible outcomes = 6

$$P(A \mid B) = \frac{1}{6}$$

Baye's Theorem



Prior Probability

Likelihood

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

Posterior Probability

Marginal Likelihood

Examples

- Pick a random card in a card pack. You know it is a diamond. What is the probability of getting a queen?

$$P\left({Queen}/{Diamond}\right) = \frac{P\left({Diamond}/{Queen}\right) \times P(Queen)}{P(Diamond)}$$

$$= \frac{1/4 \times 1/13}{1/4} = \frac{1}{13}$$

Total Probability Theorem

- Let A_1, A_2...,A_n be disjoint events that form a partition of the sample space and assume that P(Ai) > 0, ∀i = 1, 2, ..., n.
- Then, for any event B,

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + ... + P(A_n \cap B)$$
$$P(B) = P(A_1)P(B|A_1) + ... + P(A_n)P(B|A_n)$$

Total Probability Theorem

- Combining with the Baye's Theorem

$$P(B) > 0$$

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)}$$

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(A_1)P(B|A_1)+\ldots+P(A_n)P(B|A_n)}$$

Examples

- A factory produces two types of widgets: Type A and Type B. Type A widgets are known to be defective 5% of the time, while Type B widgets are defective 10% of the time. The factory produces 60% Type A widgets and 40% Type B widgets. If a randomly chosen widget is found to be defective, what is the probability that it is a Type A widget?

Examples

- A be the event that the widget is Type A.
- B be the event that the widget is Type B.
- D be the event that the widget is defective.
- We know,
  - *P(A)=0.60*
  - $P(B)$=0.40
  - $P(D|A)$=0.05
  - $P(D|B)$=0.10

Examples

- $P(A|D) = \frac{P(D|A) \times P(A)}{P(D)}$
- And from the Total Probability Theorem,
  - $P(D) = P(\ D \mid A\ ) \cdot P(A) + P(\ D \mid B\ ) \cdot P(B)$
  - *P(D)=(0.05⋅0.60)+(0.10⋅0.40)=0.07*

- *Now,*
  - $P(A|D) = \frac{0.05 \times 0.6}{0.07} = 0.43$

Probability of Independent Events

- Two events A and B are said to be independent if the probability of both events occurring together is the product of their individual probabilities.
- Mathematically, events A and B are independent if:

$$P(A \cap B) = P(A) \cdot P(B)$$

- If $A1, A2, \ldots, An$ are independent events, then the probability of all of them occurring together is:

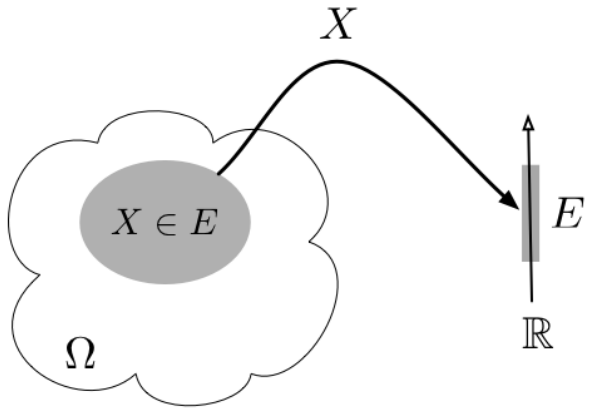
$$P(A1 \cap A2 \cap \cdots \cap An) = P(A1) \cdot P(A2) \cdot \cdots \cdot P(An)$$

Examples

- You have two fair six-sided dice. What is the probability that the first die shows a 4 and the second die shows a 5.

Random Variables

- A random variable is a function that associated a unique numerical value with every outcome of an experiment.
- Consider a random experiment with the sample space S. A random variable X(r) is a function that assigns a real number to each outcome (r) on S.
- Usually a single letter X is used for X(r) to represent a random variable.



- Random variables can be discrete or continuous.

Random Variables

- Ex:- An experiment for tossing a single coin

$$S = \{H, T\}$$

- X random variable can be defined as,
- $X = Number\ of\ heads\ obtained\ (0, 1)$

Random Variables

- Ex:- An experiment consists of tossing two fair coins simultaneously

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

- X random variable can be defined as,
- $X = Number\ of\ heads\ obtained\ (0, 1, 2)$

Random Variables

- Ex:- An experiment for rolling a fair die

$$S = \{1,2,3,4,5,6\}$$

- X random variable can be defined as,
- $X = Number\ obtained\ on\ the\ top\ (1,2,3,4,5,6)$

Random Variables

- Ex:- An experiment for finding the height of teenagers in Sri Lanka (In cm). Assume that the minimum height obtained is 100 cm and the maximum height obtained is 190 cm in the study.

$$S = \{100 \leq x \leq 190\}$$

- X random variable can be defined as,
- $X = A\ height\ of\ a\ teenager\ (Any\ value\ from\ 100\ to\ 190)$

Probability Associated with Random Variables

- Using the values of random variables, associated probabilities can be found.
- Ex:- An experiment consists of tossing two fair coins simultaneously

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

- X random variable can be defined as,

$$X = Number\ of\ heads\ obtained\ (0, 1, 2)$$

$$P(x = 1) = \frac{2}{4} = \frac{1}{2}$$

$$P(x \geq 1) = P(x = 1) + P(x = 2) = \frac{3}{4}$$

Probability Associated with Random Variables

- Ex:- Ex:- There are 5 red, 7 blue and 10 green balls in a bag. Three balls are taken randomly.
- X random variable can be defined as,

$$X = Number\ green\ balls\ taken\ (0,1,2,3)$$

$$P(x = 2) = ?$$

$$P(x \geq 2) = ?$$

Probability Associated with Random Variables

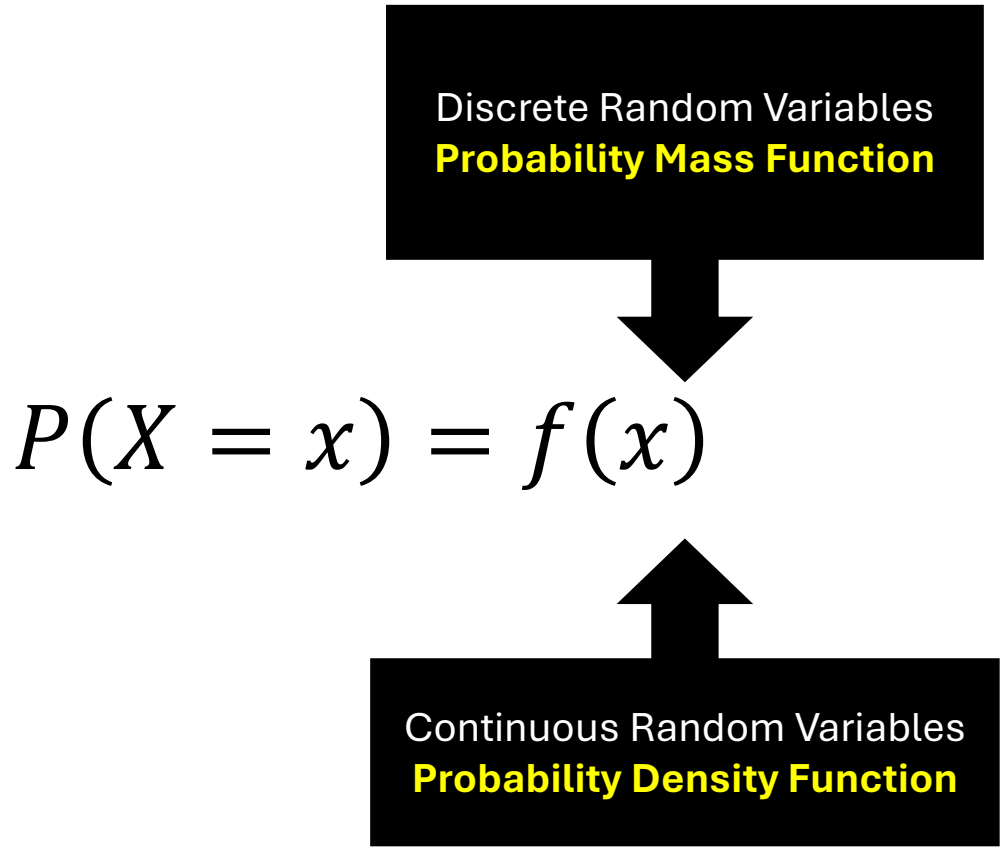
RRR	BBB	GGG
RRG	RRB	BBR
BBG	GGR	GGB
RBB	RGG	BRR
BGG	GRR	GBB
RBR	RGR	BRB
BGB	GRG	GBG
RBG	RGB	BRG
BGR	GRB	GBR

$$P(x = 2) = \frac{6}{27} = \frac{2}{9}$$

$$P(x \geq 2) = \frac{7}{27}$$

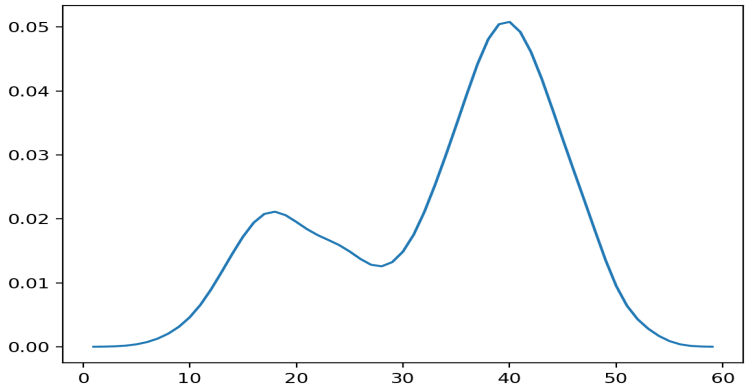
Probability Associated with Random Variables

- Probability of a random variable with any value can be represented in a functional form.



Probability Distribution

- Any function has a distribution or an arrangement with its inputs and outputs.



- In a random variable, this arrangement is called as a Probability Distribution.
- With the nature of the random variable, the distribution can be a Discrete Probability Distribution or a Continuous Probability Distribution.

Probability Distribution

- When the random variable is a discrete random variable, the distribution is a discrete probability distribution.
- There are several discrete distributions with the nature of the random variable. (Ex:- Uniform, Binomial, Poisson
- When the random variable is a continuous random variable, the distribution is a continuous probability distribution.
- There are several continuous distributions with the nature of the random variable. (Ex:- Exponential, Normal)

Moments of Discrete Probability Distribution

- Let X be a random variable.
- The mean/ expected value of X, denoted by μ_x or E(X), is defined by,

$$E(X) = \sum_x P_X(x) \ .x$$

- Let X be a random variable and let μ_x be E(X).
- The variance of X, denoted by σ_x^2 or V(X) is defined by,

$$V(X) = \sum_x (x - \mu_X)^2 P_X(x)$$

Moments of Continuous Probability Distribution

- Let X be a random variable.
- The mean/ expected value of X, denoted by μ_x or E(X), is defined by,

$$E(X) = \int_{-\infty}^{\infty} x.f_x(x)\,dx$$

- Let X be a random variable and let μ_x be E(X).
- The variance of X, denoted by σ_x^2 or V(X) is defined by,

$$V(X) = \int_{-\infty}^{\infty} (x-\mu_x)^2.f_x(x)\,dx$$

Properties of Expected Value E(X)

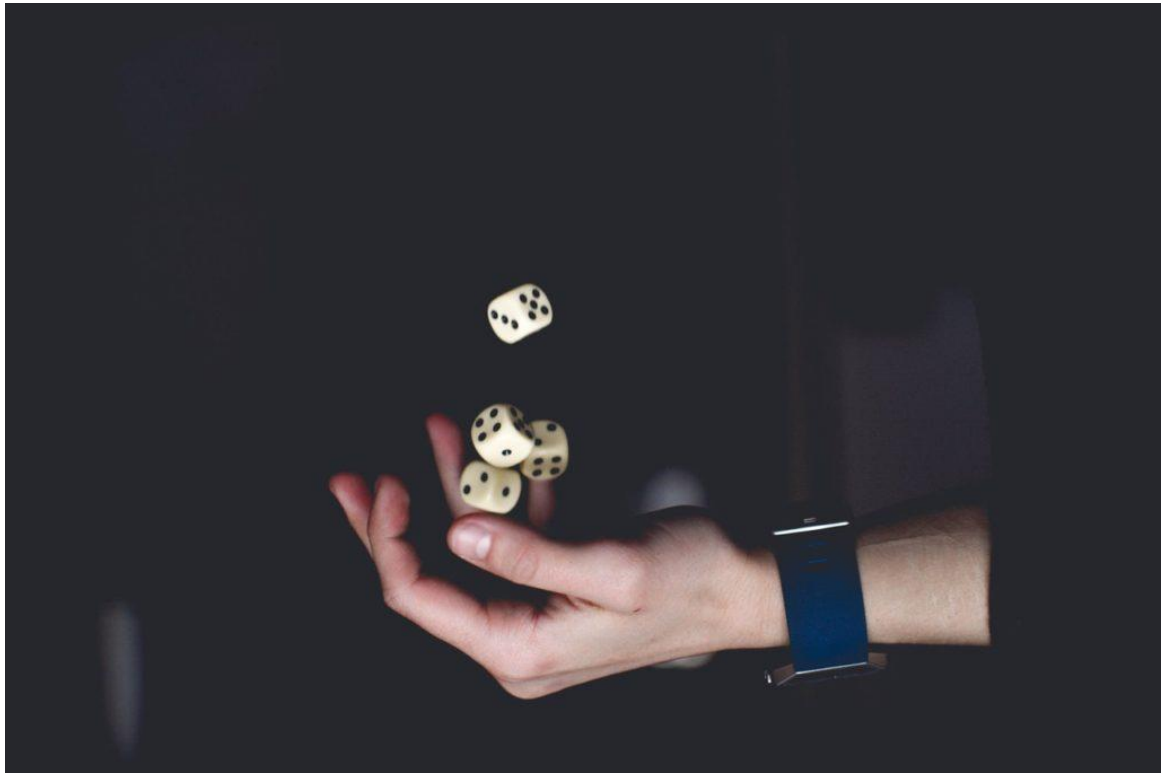
Property	Formula	Meaning
Constant	$E(c)=c$	Expected value of a constant is the constant
Scaling	$E(aX)=aE(X)$	A constant can be taken outside
Adding a constant	$E(X+c)=E(X)+c$	Adding (c) increases the mean by (c)
Linearity	$E(aX+b)=aE(X)+b$	Expected value is linear
Sum	$E(X+Y)=E(X)+E(Y)$	Works even if (X,Y) are dependent
Difference	$E(X-Y)=E(X)-E(Y)$	Expectations can be subtracted
Product	$E(XY)=E(X).E(Y)$	**Only if (X,Y) are independent**

Properties of Variance V(X)

Property	Formula	Meaning
Constant	Var(c)=0	A constant has no variation
Adding a constant	Var(X+c)=Var(X)	Shifting all values does not change spread
Scaling	Var(aX)=a^2Var(X)	Scaling by (a) scales variance by a^2
Linear transformation	Var(aX+b)=a^2Var(X)	(b) does not affect variance
Basic formula	Var(X)=E(X^2)-$[E(X)]^2$	Useful for calculating variance
Sum	Var(X+Y)=Var(X)+Var(Y)+2Cov(X,Y)	General case
Difference	Var(X-Y)=Var(X)+Var(Y)-2Cov(X,Y)	General case
Independent variables	Var(X+Y)=Var(X)+Var(Y)	If (X,Y) are independent

Introduction to Discrete Probability Distributions

- A discrete probability distribution counts occurrences that have countable or finite outcomes.



Discrete Uniform Distribution

- A random variable is defined to have a discrete uniform distribution, if the probability mass function of X is given by,

$$P_X(x) = \begin{cases} \frac{1}{N}, & \text{for } i = 1, 2, \ldots, N \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = \frac{N+1}{2} \text{ and } V(X) = \frac{N^2-1}{12}$$

Discrete Uniform Distribution

- Example: Let X represent a random variable taking on the possible values of 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, and each possible value has equal probability.
- This is a discrete uniform distribution and the probability for each of the 10 possible value is

$$P(X = x_i) = f(x_i) = \frac{1}{10} = 0.10$$

Bernoulli Distribution

- Consider the toss of a coin, which comes up a head with probability p, and a tail with probability 1-p. Define

$$X = \begin{cases} 1, & \text{if } \textit{Head} \\ 0, & \text{if } \textit{Tail} \end{cases}$$

- The random variable X is said to have a Bernoulli distribution and the probability mass function of X is given by,

$$P_X(x) = \begin{cases} P^x(1-p)^{1-x}, & \text{for } x=0 \text{ or } 1 \\ 0, & \text{otherwise} \end{cases}$$

Also, $E(X) = p$ and $V(X) = p(1-p)$

Binomial Distribution

- A coin is tossed n times. At each toss, the coin comes up a head with probability p, and a tail with probability 1-p, independently of prior tosses.
- Let X be the number of heads in the n-toss sequence.
- The random variable X is said to have a Binomial distribution with parameters n and p and the probability mass function of X is given by,

$$P_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{(n-x)}, & \text{for } x = 0, 1, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

Also, $E(X) = np$ and $V(X) = np(1-p)$

Binomial Distribution

- Example: A die is tossed 3 times. What is the probability of
  - No fives turning up?
  - 1 five?
  - 3 five?

Binomial Distribution

This is a **binomial** distribution because there are only 2 possible outcomes (we get a 5 or we don't).

Now, $n = 3$ for each part. Let $X =$ number of fives appearing.

(a) Here, $x = 0$.

$$P(X = 0) = C_x^n p^x q^{n-x} = C_0^3 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^3 = \frac{125}{216} = 0.5787$$

(b) Here, $x = 1$.

$$P(X = 1) = C_x^n p^x q^{n-x} = C_1^3 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = \frac{75}{216} = 0.34722$$

(c) Here, $x = 3$.

$$P(X = 3) = C_x^n p^x q^{n-x} = C_3^3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0 = \frac{1}{216} = 4.6296 \times 10^{-3}$$

Poisson Distributions

- The number of events occurring (X) within a fixed period of time has Poisson distribution and its probability mass function is given by

$$P_X(x) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!}, & \text{for } x = 0, 1, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = \lambda \text{ and } V(X) = \lambda$$

- Poisson distribution can be effectively used to approximate Binomial probabilities when the number of trials n is large, and the probability of success p is small.

Poisson Distributions

- Example: A life insurance salesman sells on the average 3 life insurance policies per week. Use Poisson's law to calculate the probability that in a given week he will sell
  - Some policies
  - 2 or more policies but less than 5 policies.

Poisson Distributions

Here, $\mu = 3$

(a) "Some policies" means "1 or more policies". We can work this out by finding 1 minus the "zero policies" probability:

$$P(X > 0) = 1 - P(x_0)$$

Now $P(X) = \frac{e^{-\mu}\mu^x}{x!}$ so $P(x_0) = \frac{e^{-3}3^0}{0!} = 4.9787 \times 10^{-2}$

Therefore the probability of 1 or more policies is given by:

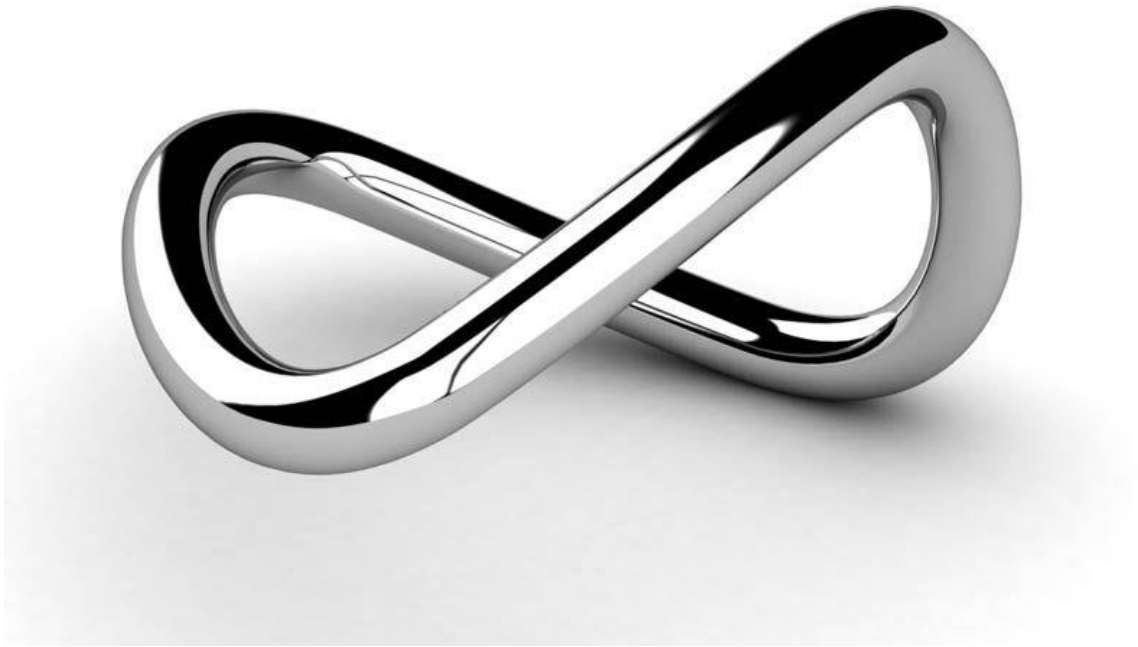
$$\begin{aligned}\text{Probability} &= P(X \geq 0) \\ &= 1 - P(x_0) \\ &= 1 - 4.9787 \times 10^{-2} \\ &= 0.95021\end{aligned}$$

(b) The probability of selling 2 or more, but less than 5 policies is:

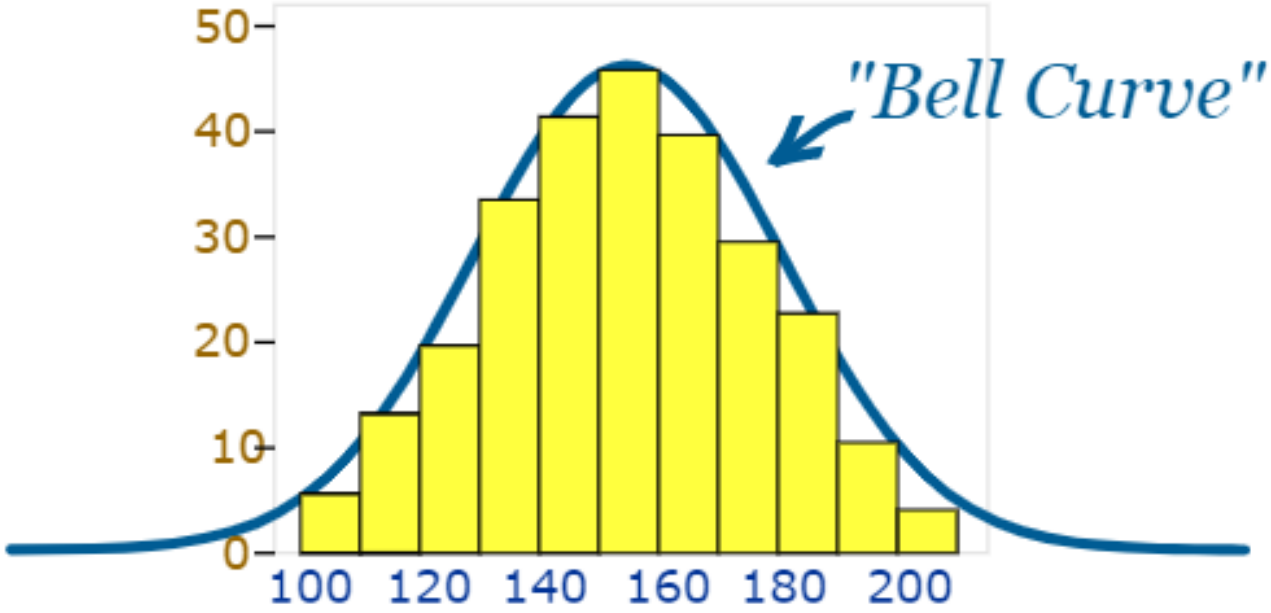
$$\begin{aligned}P(2 \leq X < 5) \\ &= P(x_2) + P(x_3) + P(x_4) \\ &= \frac{e^{-3}3^2}{2!} + \frac{e^{-3}3^3}{3!} + \frac{e^{-3}3^4}{4!} \\ &= 0.61611\end{aligned}$$

Introduction to Continuous Probability Distributions

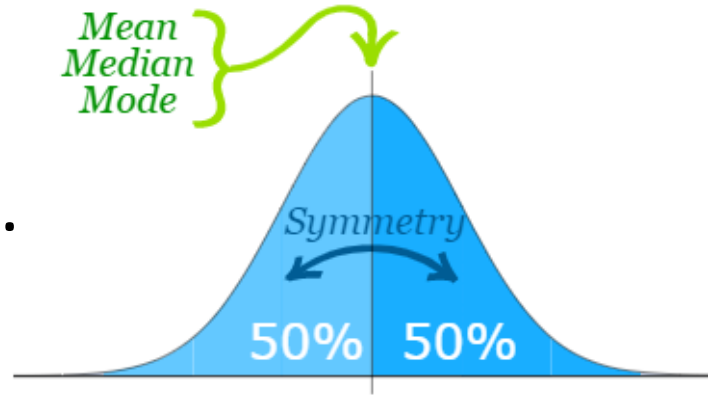
- A discrete probability distribution counts occurrences that have infinite outcomes where outcomes can fall anywhere on a continuum.



Normal Distribution



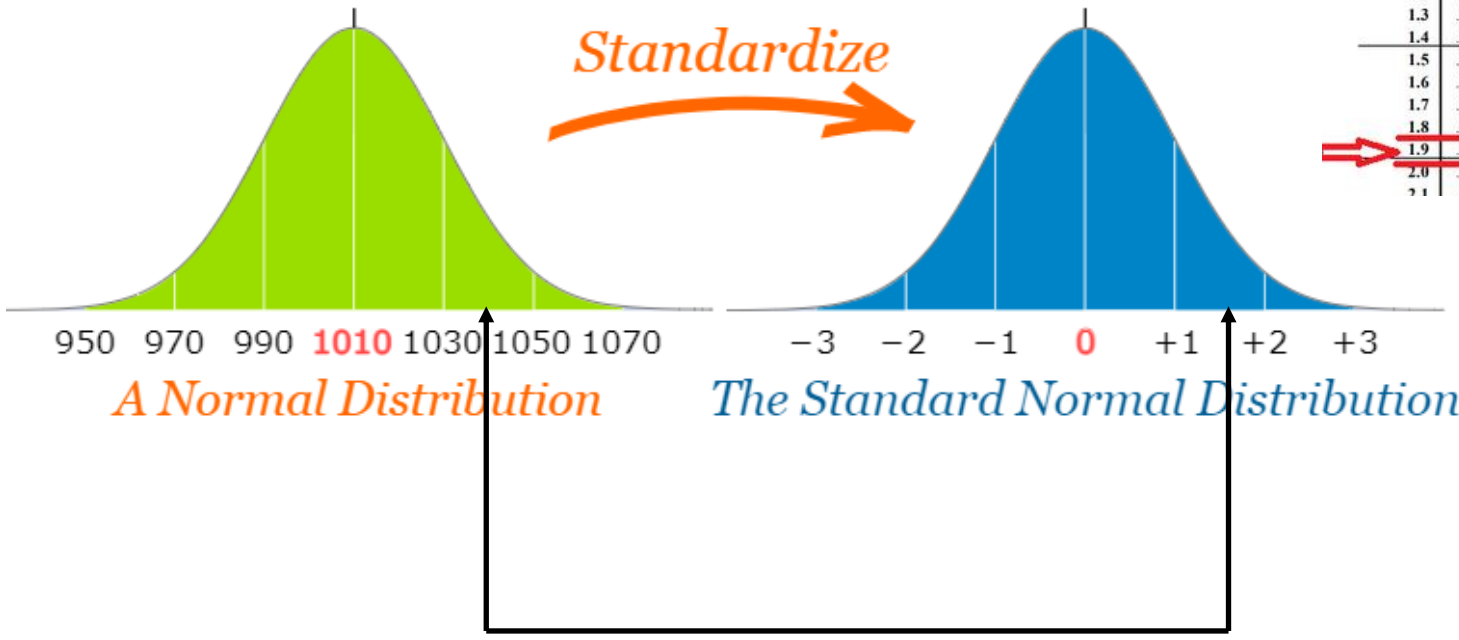
$X \sim N(\mu, \sigma^2)$



Important parameters are the **mean**$(\boldsymbol{\mu})$ and the **variance** $\left(\boldsymbol{\sigma^2}\right)$**.**

Standard deviation $(\boldsymbol{\sigma})$ is the square root of variance.

Standard Normal Distribution



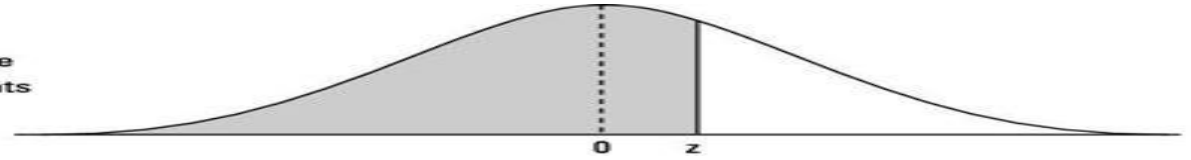
STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574

$$z = \frac{x - \mu}{\sigma}$$

Standard Normal Distribution – Z Table

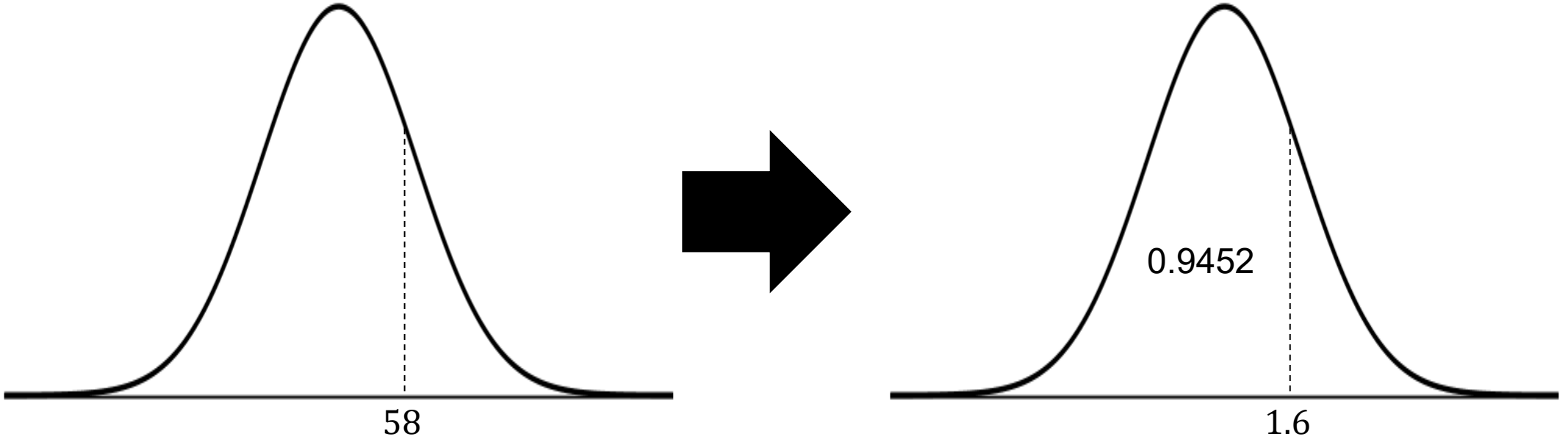
Number in the table represents $P(Z \leq z)$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998
3.6	.9998	.9998	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999

Example

Mean age of a population is 50. If the standard deviation is 5, what proportion of people are in the population from age 40 to age 60. What proportion of people are in the population from age 42 to age 58?



Example - Continue

What proportion of people are in the population from age 42 to age 58?

- Find the proportion of people below age 58
- Find the proportion of people below age 42
- Get the difference between these proportions

What proportion of people in the population who are older than 58?

Exponential Distribution

- A random variable is defined to have a exponential distribution, if the probability density function of X is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{for } 0 \leq x \leq \infty \end{cases}$$

Also, $E(X) = \frac{1}{\lambda}$ and $V(X) = \frac{1}{\lambda^2}$

Example

- Example: At a certain location on a highway, the number of cars exceeding the allowed speed limit per half hours is 8.4. What is the probability of waiting time is 5 minutes?

$$f(X = 5\ minutes) = 0.28e^{-0.28\times5} = 0.07$$

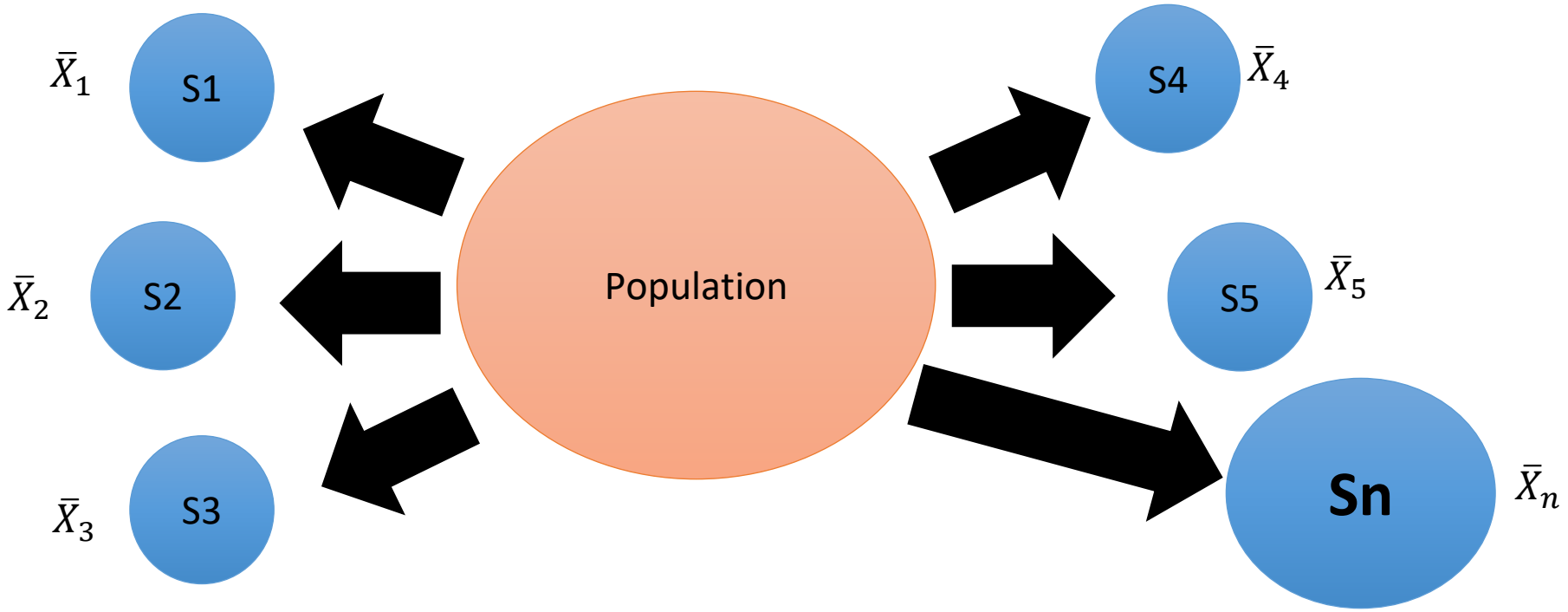
- What is the probability of waiting time less than 5 minutes?

$$f(X < 5\ minutes) = \int_0^5 0.28e^{-0.28x}\ dx$$

Sampling Distribution

- In a population any number of samples can be selected.
- From each sample any statistic can be calculated.
- All these statistics are estimates for the population parameter interested.
- Consider that we want to estimate the population mean of a population.
- Then we can get many samples from that population and sample means can be calculated.

Sampling Distribution



Sampling Distribution

- Sample statistics are random variables.
- From sample to sample they vary.
- As a result, sample statistics also have a distribution called the sampling distribution.
- These sampling distributions, similar to distributions discussed previously, have a mean and standard deviation.
- Standard deviation of a sampling distribution is called as standard error.

Sampling Distribution of the Sample Mean

- If $x_1, x_2, x_3, \ldots, x_n$ is a random sample of size n taken from a normal population with mean μ and variance σ^2, then the distribution of sample mean $(\bar{X})$ is also normally distributed with mean μ and variance $\frac{\sigma^2}{n}$.
- **Central Limit Theorem -** Even though the sample is taken from a population which is not distributed normally, when the sample size increases, the sample mean $(\bar{X})$ is also normally distributed with mean μ and variance $\frac{\sigma^2}{n}$.